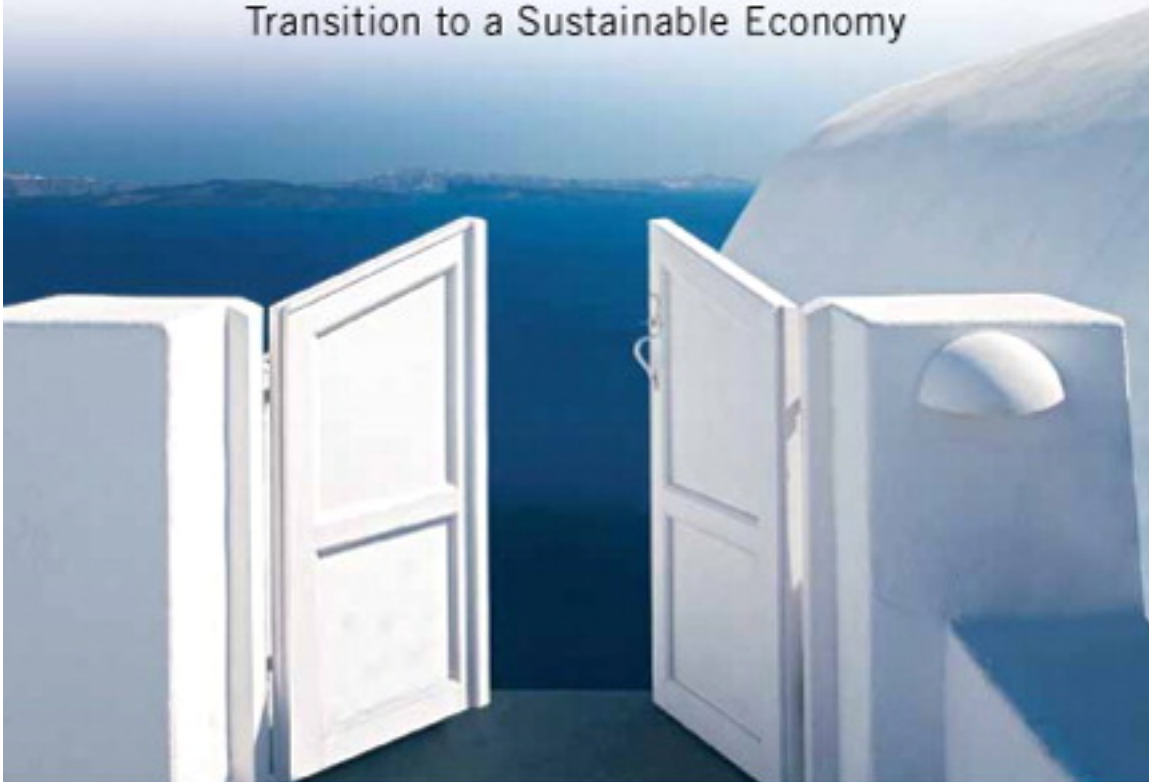




Sustainable Venturing

Entrepreneurial Opportunity in the
Transition to a Sustainable Economy



Thomas J. Dean

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CHAPTER

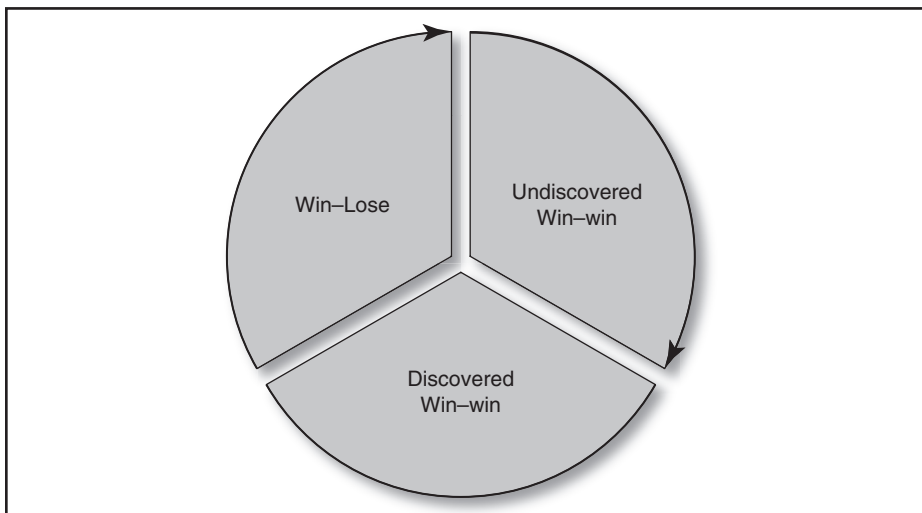
SYSTEMS AND OPPORTUNITIES

INTRODUCTION

In the first chapter, we learned that economic opportunities either exist within the system of rewards present in the economy at a given time or can be created through changes in the system of rewards within a reasonable period. In other words, the nature of incentives present or possible within an economy drives opportunities. Stated differently, the economic behaviors of individuals, corporations, and entrepreneurs are driven by the systems that reward them for various actions. The character of these rewards varies greatly and ranges from the prices of goods, to legal constraints, to social norms that dissuade certain behaviors or encourage others. The system of incentives (and disincentives) is generally referred as the *economic system*. This chapter considers a number of important points about economic systems, sustainability, and entrepreneurship.

First, it is useful to consider the underlying goal of an economic system and how we might gauge the difference between a good economic system and a bad one. Many think that the goal of an economic system is to make money and consider wealth generation as a proxy for success within an economy. But the generation of monetary wealth is at best an intermediate goal of economics. Money itself is no more useful than the paper on which it is printed. Money's potential translation into value is what matters. *The fundamental goal of an economic system is not to make money per se, but to provide value to members of society.* Economists refer to this as the *satisfaction of needs and wants*. The ultimate test of an economic system is the extent to which that system is providing value for members of the society that it serves. Value is also a humanly determined

Figure 2.1 divides the potential social and environmental activities of business into three categories. The first are *win-win* activities, which have already been discovered and implemented by businesses (*discovered win-win actions*) and offer private economic benefits and better social or environmental outcomes. These conditions represent potential opportunities but are relatively less interesting to the sustainable entrepreneur because they are already in widespread

FIGURE 2.1 Win-Win and Win-Lose Conditions

use and have little potential for superior economic return. The second category includes *win-win* conditions that are undiscovered or underexploited (*undiscovered win-win actions*). These conditions represent substantive opportunities for expanding actions that simultaneously reduce degradation or enhance social or environmental resources while offering the potential for superior profitability. Many of these are cataloged in popular books on sustainable enterprise and the subject of much discussion in later parts of this text. The third are *win-lose* conditions, which might offer the opportunity to enhance environmental outcomes, but are not incented by current market conditions. From an environmental and entrepreneurial perspective, these activities are perhaps the most intractable ones. But because they are the toughest nuts to crack, so to speak, they may offer the greatest economic returns, if the market conditions on which they are based are changing or may be altered by entrepreneurial action. The transformation of win-lose conditions to win-win conditions is of great interest to the environmental entrepreneur. The ~~solution to~~ win-lose conditions rests in changes to the market institutions that create economic incentives.³ Understanding the nature of these institutions is the topic of the remainder of this chapter and the next.

ECONOMIC SYSTEMS AND INSTITUTIONS

Economic systems exist within, and are a fundamental part of, broader social systems. Stated differently, *economic systems are embedded within social systems* and the associated norms, values, beliefs, priorities, and legal structures of those

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social systems. This helps account for the vast differences in economic systems throughout the world. Although some tend toward a greater degree of hierarchical management associated with socialism, others tend toward a more free-market orientation. Regardless of a system's economic philosophy (e.g., socialist or capitalist), systems may also vary in the specific rules and rights that govern and drive competitive and corporate behavior. Whether it is tax policies, rules of incorporation, or environmental regulation, companies in different societies face different sets of incentives and constraints. Economic systems are embedded within social systems in the sense that they are inherently driven by the nature of social norms, values, beliefs, and priorities. This is important to entrepreneurs because, to the extent that social systems drive the economic systems of rewards, entrepreneurial opportunity will be driven by the priorities of social systems.

ECONOMIC INSTITUTIONS

The term used to broadly describe the drivers of incentives and rewards in society is *economic institutions*. Institutions are “the humanly devised constraints that shape human interaction.”⁴ They are argued to “structure incentives in human exchange, whether political, social, or economic”⁵ and therefore serve as the central driver of economic activity and development.⁶ They are, in essence, the rules of the game by which economic actors play. Institutions are not, as most people think, specific organizations, but rather the underlying structures on which those organizations are founded. Primary examples of economic institutions that are important to entrepreneurs include the granting of legal status to corporations, the system of property rights by which individuals own land and products, and the legal frameworks of intellectual property protection (like patents). Institutions can range from highly formalized structures, such as laws, to informal customs and values, such as a handshake agreement or cultural acceptance of innovative behavior. Both formal and informal institutions govern how people and organizations act and the rewards resulting from various actions.

Patent law is probably the best example of the role of economic institutions with respect to entrepreneurs and the returns to innovation. Patent laws vary from country to country, but generally speaking, patent laws grant temporary exclusive right to commercial use of a novel product or technology. Patent laws allow holders of patents to sue companies that infringe on their intellectual property, and such laws are important to economic development and social value creation. By granting exclusive rights to innovators, society increases entrepreneurs' potential return to innovation and society gains from increases in the rate of innovation. As a result, the institution of patent law helps drive innovation and entrepreneurship.

Formal rights to real estate are another important example of the role of economic institutions. A system of legal rights to land ownership and use creates incentives to invest in assets on property because it secures future ownership of those investments. Moreover, in most nations individuals can borrow capital against their real assets and employ that capital to create additional value in

society. In the absence of such property rights, land stewardship can suffer, and individuals have very few means of unlocking opportunities to borrow capital against their real assets, further perpetuating the cycle of poverty. In the book *The Mystery of Capital*, Hernando de Soto argues that the lack of formal property rights for land in underdeveloped countries has been one of the primary reasons that those countries have been unable to emerge from poverty.⁷ DeSoto also emphasizes the absence of individual legal identity, bureaucratic barriers to company formation, and other institutional failures that prevent the accumulation of capital and inhibit entrepreneurial behavior. Because of the tremendous impact economic institutions have on society's ability to generate wealth, they are indeed a major consideration for entrepreneurs seeking opportunities in various political regimes and economic systems.

Economic historian Douglass North was awarded the Nobel Prize in economics for his work on the link between economic institutions and economic development.⁸ He argues that economic development in Western economies over the past centuries resulted directly from the development of new institutions that encouraged productive entrepreneurial endeavors. North emphasizes how new institutions develop over time, particularly in the long-term. Again, patent law is an instructive example, as it shows how the development of economic institutions effect economic development over the long run. The history of patents goes back as far as ancient Greece, but the first formal patent law was not put into place until the 1400s in the Republic of Venice.

Thus, environmental entrepreneurship rests on the principles that *economic systems and institutions are in a continual state of change* and that *the manner in which economic institutions are evolving is at least partially in response to the emerging needs of society and the increasing scarcity of resources*. As Douglass North and Robert Thomas explain,

Economic institutions, and specifically property rights, are generally considered by economists as parameters; but for the study of long-term economic growth, they are clearly variables, historically subject to fundamental change.⁹

Economic institutions need not be laws and regulations monitored and enforced by the government. Institutions can also be cultural, tied to the social norms of the community. For example, certain tasks or jobs may only be restricted to those of a particular gender, race, or class. Religion is another important institution that influences the way in which a society conducts business. Knowing and understanding how different societies and cultures operate is imperative to the sustainable entrepreneur. Even if the endeavor is completely legal, yields a positive impact, and can generate profit, if it goes against a deep seeded cultural institution, it may never get off the ground because the community cannot or will not support it.

ECO-INSTITUTIONAL CONVERGENCE

Eco-institutional convergence is the idea that economic institutions are evolving in response to fundamental ecological challenges and that, as a result,

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entrepreneurs will be increasingly incented to resolve these ecological challenges. In other words, the nature of economic institutions will evolve and converge with the need for the resolution of global ecological challenges. But most important, as these changes alter the rules of the game in competitive markets, they create economic opportunity for businesses that create sustainable economic and environmental value.

New institutions driving market opportunities in environmental entrepreneurship have been emerging rapidly in the past decades. They range from voluntary industry labeling initiatives, to international policy agreements, to new market mechanisms for trading pollution credits. Such developments need not be driven by government legislation. Individuals or groups with an interest in environmental sustainability or economic profit may also create new institutions in a voluntary manner. Institutional changes driven by legislation also need not be regulatory, as a host of other policies drive economic behavior. Alternatives to command-and-control regulations include new property rights, certification and labeling schemes, disclosure requirements, deposit-refund schemes, new market mechanisms, pollution taxes, and alternative subsidies. Finally, eco-institutional convergence may be driven as much from the bottom-up as it is from the top-down.

Further understanding of the nature of eco-institutional convergence and the opportunities created by this convergence can be gained by understanding current and historical imperfections in our market institutions, as those imperfections are seen as largely responsible for our environmental challenges.

MARKET FAILURES

The field of *environmental economics* is concerned with whether market systems are effective and efficient in allocating environmental resources, and if not, what might be done about it. Environmental economists approach *environmental challenges primarily as an economic problem*. In other words, they trace the source of environmental challenges back to the economic system and ask how the economic system has resulted in our modern environmental problems and/or how it might solve them. The concept of *market failure* is one of the most prevalent concepts in the field, as market failure is largely viewed as the economic cause of environmental degradation. Market failure can be viewed as the *inability of a system of market institutions to incent desirable activities or impede undesirable ones*.¹⁰ Market failures are, in essence, conditions that drive entrepreneurs and businesses to make win-lose decisions (win for the company, but lose for society). They are called *market failures* because the system incents a suboptimal outcome. In other words, in such situations a corporation or entrepreneur pursues its own self-interest to the detriment to society at large. The market is failing to provide the most effective or efficient solution.

Market failures that have negative impacts on environmental resources have been termed *environmentally relevant market failures*.¹¹ Environmentally

Market Failures 23 ■

relevant market failures are particularly prevalent because many environmental resources, such as the atmosphere, have characteristics that make them less amenable to market allocation.¹² Air and water, for example, can be difficult resources to manage through market systems because they tend to be transient and indivisible.¹³ It is also the case that many of these resources have been abundant in the past, but are just now becoming scarce. As scarcity increases, the inability of our market system to effectively value and steward environmental and natural resources has become more obvious and more problematic. This condition is popularly referred to as *the tragedy of the commons*, a situation in which individuals collectively engage in depleting a shared resource because each individual is acting in his or her own best interest in the short term, even though it is in no one's best interest in the long run.¹⁴

The existence of environmentally relevant market failures detracts from the sustainability of an economic system by allowing the degradation of the natural assets that are necessary to generate future economic value.¹⁵ One way to understand market failures is through the concepts of *social costs* and *private costs*.¹⁶ Social costs represent all of the costs of an activity incurred by the entirety of a society. They are typically contrasted with *private costs*, which are the costs borne by the individual or organization involved in an activity or transaction. Where social costs are greater than private costs, the individual or organization performing an activity does not bear all of the costs of that activity. For example, when coal is burned to create electricity, emissions containing particulates, sulfur dioxides, carbon dioxides, and mercury are released into the atmosphere. Unless required to eliminate these emissions or otherwise compensate society, the electric utility does not bear the costs of the emissions. But society does, because those emissions affect health, agricultural productivity, and ecosystems downwind of the utility. Such costs are often referred to as *external costs*, because they exist external to the party that created them, which does not pay for them.

From a systems perspective, the presence of social costs above and beyond private costs (often referred to as *external costs* or *negative externalities*) creates perverse market incentives that are of particular concern for the efficient functioning of an economic system, and by extension the proper allocation of resources. If, for example, utilities do not pay the full costs of production from coal, the dynamics of supply and demand suggest that society ends up with higher production of electricity from coal than might otherwise be ideal. Where substantial external costs of an activity exist, an industry or a producer is in essence "subsidized" because it does not have to pay the full costs of that productive activity. From an entrepreneurial perspective, the ability to externalize costs on others makes the production of electricity from coal relatively more attractive than the production of electricity from other energy sources, such as wind or solar power, because the electricity entrepreneur choosing to use coal does not have to pay the full costs of power generation.

Market failures may also occur from the creation of *social benefits* that do not accrue to the individual or organization engaged in that activity. One

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classic example of external benefit occurs in vaccinations. Social benefits from an individual receiving a vaccination for a communicable disease accrue to others in society because the vaccination reduces the probability of others getting the disease (in addition to reducing the probability of the individual acquiring the disease). Because individuals are not necessarily willing to pay to prevent others from getting a disease, it is often suggested that the market system will result in fewer vaccinations than what would be ideal from society's perspective. Social benefits above private benefits are referred to as *external benefits* or *positive externalities*. Industries or companies producing positive external benefits for which they are not rewarded will tend to be relatively less incented to produce their product or service. Most important, an entrepreneur who is unable to capture all of the benefits of a product or service is less likely to offer that product or service. So the entrepreneur must find a way through the current system of rewards, or by altering the system of rewards, to capitalize on all of the benefits they create for society.

The ability of an entrepreneur to *appropriate* the returns from the benefits created by its activity is important in the effective functioning of a market system, as well as to the viability of an entrepreneurial endeavor. Likewise, the ability to require producers to pay the full costs of their activities is important for market systems and for the ability of entrepreneurs with environmentally superior business models and technologies to be successful. If external costs and benefits are not borne by the producer of goods or services, the system of rewards may be incenting suboptimal economic performance. In other words, there are dormant opportunities waiting to be exploited by the entrepreneur who can find a way to successfully navigate the system.

The *polluter pays principle*, supported by the Organization for Economic Cooperation and Development, states that polluters should bear the full costs of any damage caused by their production of goods and services. If the full costs of damages to society are integrated into the production of pollution-generating goods or services, the prices to consumers may rise, making cleaner substitutes more economically viable and attractive to the consumer.

Consistent with the polluter pays principle, *full-cost pricing* exists when prices reflect all of the costs of production, including the costs of environmental degradation and negative social impacts. Full-cost pricing enables sustainable development through a free-market system because prices are a more accurate reflection of the resources necessary to produce a given product or service. In such a scenario, environmentally or socially harmful activities look relatively less attractive in the marketplace than those that do not create such damages, because customers face a relatively lower price for the less damaging products or services. The goal, then, from a systems perspective, is that market failures remain relatively minimal, and private costs and benefits be approximate to social costs and benefits. While there are substantive efforts to achieve full-cost pricing in our economic system, most observers suggest that prices still do not reflect many of the social and environmental damages of some productive activities.

MARKET FAILURE AS ENTREPRENEURIAL OPPORTUNITY

Understanding market failure is central to understanding the potential economic opportunities available in environmental entrepreneurship for two reasons. First, the magnitude of *potential* opportunities for sustainable venturing corresponds to the level of market failure and to the value being lost to society because of the degradation of environmental resources. To the extent that environmental degradation results in diminished ecosystem services, social value can be obtained by preserving or enhancing the resources affected, meaning that entrepreneurs who can capture some of the value created through the preservation or betterment of the natural resources can start profitable ventures. We are beginning to see this phenomenon occur in and around many ecosystem services. Watershed protection is one that is of particular importance in the western United States. Private companies and governments are increasingly willing to pay for the availability of clean water from a natural source (e.g., streams, rivers, or lakes). By paying someone to protect the watershed from which the water is generated, the company is better off, and the entity providing the service of maintaining the watershed has a revenue-generating opportunity. When considering the level of degradation in various ecosystems, the value that can be captured by such activities can be substantial.

Second, to the degree that we can understand the market failures preventing the efficient use of environmental resources, we can gauge the means by which entrepreneurs might resolve those failures and help preserve environmental resources. Biofuels can replace fossil fuels with less harmful alternatives. Biofuels become more economically viable when consumers of fossil fuels are required to pay the external costs of burning them. By working with the institutional framework to make such a switch economically viable for both the consumer and the producer, an entrepreneur can put resources to greater and better use while preserving natural resources and promoting environmental stewardship. Understanding how to build and/or exploit opportunities in eco-institutional convergence is a central strategy of sustainable venturing.

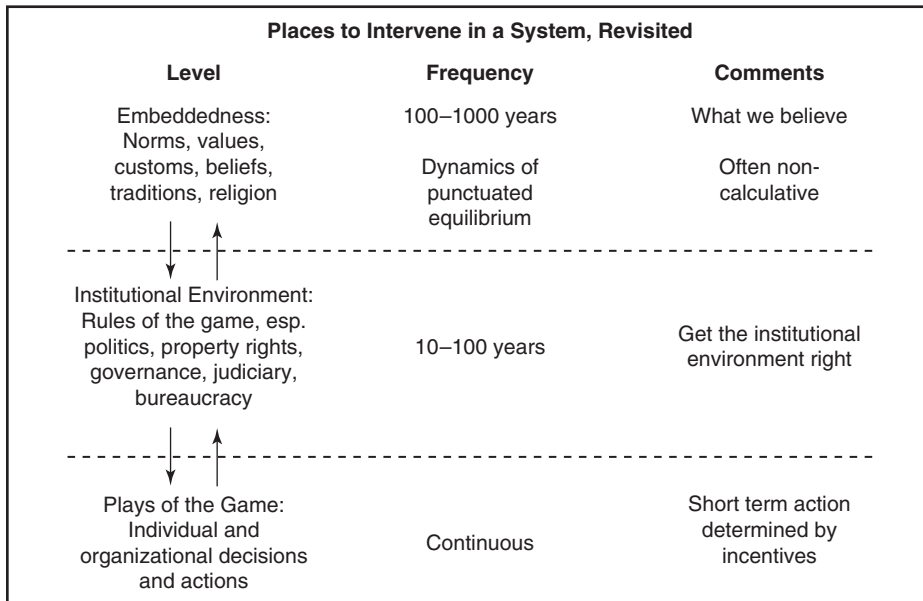
It is important to note that many market failures do not permanently exist within the economic system, nor does the market create them, per se; in other words, the term *market failure* is somewhat misleading. Many market failures result from the regulatory or legal environment put in place by the institutional actors within the economic system, while others are a result of society's norms, values, and traditions, all of which can change and evolve over time. Regardless of the reason for market failures, one should not assume that a market failure is permanent.

PLACES TO INTERVENE IN A SYSTEM

The above discussions are meant to make an important point about economic systems, sustainability, and entrepreneurial opportunity: that it is not always sufficient to take the system or the rules of the game as they are. Rather, much of sustainable

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FIGURE 2.2 Places to Intervene in a System



Source: Williamson, O. 2000. The New Institutional Economics: Staking Stock, Looking Ahead. Journal of Economic Literature. Vol XXXVII September: 595–613. D. Meadows, “Leverage Points: Places to Intervene in a System,” Solutions 1, no. 1(2009): 41–49, available online at <http://www.thesolutionsjournal.com/node/419>.

venturing is about changing the system to achieve better results and enable a new venture. System thinker and sustainable-development advocate Dana Meadows usefully summarizes this fact in her writings on “places to intervene in a system.”¹⁷ Figure 2.2 combines her approach with that of economist Oliver Williamson¹⁸ to emphasize the hierarchy of leverage points in an economic system. The figure shows three levels of a system: the *plays of the game*, the *institutional environment*, and *embeddedness*. The *plays of the game* are decisions and actions made by individuals and organizations, including entrepreneurs. The institutional environment consists of the rules of the game by which such decisions are made, including the legal structures, property rights, political systems, and other formal and informal rules that determine the payoffs (or incentives) for various decisions and actions. At the highest level is embeddedness, which consists of the norms, values, customs, and beliefs that determining the nature of the rules and associated incentive structures.

Dana Meadows has pointed to the fact that the most effective place to alter a system is at the highest level, that of the embeddedness. This is because changes to norms, values, and beliefs affect the entire system. In other words, changes in embedded norms can alter the institutional environment, which then impacts individual decisions. In short, if you want to have the greatest impact on sustainable development, change what people believe and value. Unfortunately, the time frame for changing such beliefs is long, arguably in the range

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of centuries. Fortunately, beliefs can also change rapidly, often in response to an external event or occurrence. The dynamics of periods of little change, interspersed by rapid revolutionary change, are known as *punctuated equilibrium*. The next most-influential place to intervene in a system is at the institutional level, a dynamic discussed extensively above. In short, entrepreneurs have the opportunity to change the rules of the game in a way that enhances sustainability by increasing the incentives for more sustainable business models. We'll address the specifics of such actions (institutional strategies) in the next chapter. For now, the most important point is that sustainable entrepreneurs have the opportunity both to act within existing systems and to act on existing beliefs and institutions in order to enable their ventures. The best results may be achieved by operating on the latter of these, where potential leverage in the system is at its greatest, even though such changes may be the most difficult.

CLOSING THOUGHTS: GREAT CAPITALISM IS GREAT ENVIRONMENTALISM

Many argue that the cause of our environmental challenges and ecological unsustainability is a system of markets and government policies that sacrifice our longer-term environmental and economic health for current economic gains. Sometimes it seems that business enterprises are more motivated to pollute and diminish our natural resources than to enhance, or at least sustain, them. Capitalism, for all its benefits, may be eating at the very foundation upon which it is based, the ecological systems from which social wealth and value is ultimately derived. Such conditions have caused many environmentalists to question the value of capitalism and even to suggest that it be replaced with less market-oriented economic systems. But such approaches fail to understand the capacity of markets and entrepreneurship to help resolve our environmental ills and to build a more sustainable economy. Rejecting capitalism because of our ecological challenges would certainly be shortsighted. On the other hand, ignoring our current environmental challenges and the unwanted effects of our economic activity is equally inadvisable.

The key is to make markets and capitalism work for the environment rather than against it. Such an approach can capture the powerful benefits of capitalism and entrepreneurship while helping resolve our environmental challenges. As industrialist Stephan Schmidheiny has said, "The cornerstone of sustainable development is a system of open, competitive markets in which prices are made to reflect the cost of environmental as well as other resources."¹⁹ According to this line of thought, *great capitalism is great environmentalism, and great environmentalism is great capitalism*.

Great capitalism as great environmentalism is closely aligned with the concept of *free-market environmentalism*. Free-market environmentalism emphasizes the role that properly structured markets and decentralized decision-making can play in effectively allocating natural and environmental resources.²⁰ Free-market environmentalism assumes that (1) individuals have a tendency to make decisions in their self-interest, (2) decentralized markets are better allocators of resources than centralized planning systems or political hierarchies, and

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(3) well-specified property rights for environmental and natural resources will internalize costs of production and create a system in which prices will represent full costs. Most important, free-market environmentalism emphasizes the role that properly structured market systems can play in motivating innovative entrepreneurial solutions to environmental challenges.

Yet, it is not entirely clear how far a market-based approach can go in preserving certain ecological resources. While the purpose of this book is to emphasize market solutions and the opportunities available in the restructuring of market incentives, markets are nonetheless still a human tool, and thereby serve human wants and needs. Many environmentalists see ecological resources as having value that transcends their worth to human populations, a view typically known as *ecocentrism*. According the Food and Agriculture Organization of the United Nations *environmental resources* have an intrinsic value of their own, as well as of value for the longer-term use by human populations.²¹ Yet only humans operate in markets and political systems, so free-market environmentalism and market-based approaches succeed only in so far as humans value environmental resources. Nevertheless, there is little doubt that we can go much further in aligning our market systems with the human value of ecological resources and thereby act to better sustain them.

Central to great capitalism and great environmentalism is a system of rewards that encourage ecologically sustainable entrepreneurial action. One of the most compelling management articles is Stephen Kerr's "On the Folly of Rewarding A While Hoping for B," in which the author points out the basic truism that "Whether dealing with monkeys, rats, or human beings, it is hardly controversial to state that most organisms seek information concerning what activities are rewarded, and then seek to do those things, often to the virtual exclusion of activities not rewarded."²² And so it is with entrepreneurs: if our markets do not reward our entrepreneurs for protecting and enhancing our environmental resources and natural capital, they will not do so, and it is really quite difficult to blame them for that.

Conversely, if our markets *do* reward entrepreneurs for more sustainable enterprise, they will race to the decisions and actions that enhance sustainability, and the sustainable behaviors they pursue will be replicated across the world, whether or not each entrepreneur is passionate about saving the planet. Thus, the alignment of economic opportunity and entrepreneurial self-interest with ecologically sustainable outcomes is the goal of great capitalism and great environmentalism, and this alignment is the foundation upon which environmental entrepreneurs are changing our world and will pioneer a more sustainable future.

Key Learnings

1. The ultimate goal of an economic system is to provide value for its constituents. Better economic systems provide greater value for society.
2. *Win-win* situations occur when actions that benefit an individual or organization also benefit society—that is, when self-interest is aligned

with social value creation. Win-win situations are common and one of the primary benefits of the capitalistic economic system.

3. *Win-lose* situations exist when the pursuit of self-interest comes at the expense of society. ~~Win-lose situations do exist within our economic system.~~
4. Win-win and win-lose conditions are determined by the nature of institutions in place within an economic system.
5. Institutions are the ways of doing things and rules of the game that create incentives in the economic system.
6. Institutions evolve over time, at least partially in response to emerging needs of society. *Eco-institutional convergence* refers to the way that institutions change in response to social and environmental challenges.
7. New, changing, or emerging institutions create opportunities for entrepreneurs.
8. Sustainable entrepreneurs can intervene in the system by creating new institutions that create win-win conditions and enhance sustainability. Intervening at the systems level can be difficult, but can also provide the greatest potential for impact.
9. *Market failures* are incidences in the economic system that create win-lose conditions. Many market failures occur with respect to environmental resources.
10. Market failures may be seen as potential opportunities for the sustainable entrepreneur.

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